

Task 1 (26 Points)

* a) Lagrangian function

(2 P)

$$L(\mathbf{x}, \boldsymbol{\mu}) = f(\mathbf{x}) + \sum_{i=1}^q \mu_i h_i(\mathbf{x}) = (1-s)x_1 + x_2 + \mu_1(2x_1 + x_2 - 4) - \mu_2 x_1 - \mu_3 x_2 \quad (1)$$

Optimality conditions

$$\nabla_{\mathbf{x}} L(\mathbf{x}, \boldsymbol{\mu}) = \begin{bmatrix} 1-s+2\mu_1-\mu_2 \\ -1+\mu_1-\mu_3 \end{bmatrix} = \mathbf{0} \quad (2a)$$

$$h_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, 3 \quad (2b)$$

$$\mu_i \geq 0, \quad i = 1, \dots, 3 \quad (2c)$$

$$h_i(\mathbf{x})\mu_i = 0, \quad i = 1, \dots, 3 \quad (2d)$$

* b) Consider different cases of active constraints

(10 P)

- h_1 active: $\mu_1 \geq 0, \mu_{2,3} = 0$

 From (2a) follows $\mu_1 = 1$ and

$$1-s+3=3-s=0 \Rightarrow s=3.$$

 From the active $h_1 = 2x_1 + x_2 - 4 = 0$ follows

$$x_2 = 4 - 2x_1 \text{ and since } x_2 \geq 0 \Rightarrow \mathbf{x}^* = [x_1, 4 - 2x_1]^T \text{ with } x_1 \in [0, 2].$$

- h_2 active: $\mu_2 \geq 0, \mu_{1,3} = 0$

 The second element in (2a) can not be fulfilled. $\Rightarrow$ No optimal solution.

- h_3 active: $\mu_3 \geq 0, \mu_{1,2} = 0$

 From the second element in (2a) follows $\mu_3 = -1 \not\geq 0$. $\Rightarrow$ No optimal solution.

- h_1 and h_2 active: $\mu_{1,2} \geq 0, \mu_3 = 0$

 From (2a) follows $\mu_1 = 1$ and

$$1-s+2-\mu_2=3-s-\mu_2=0 \Rightarrow s=3-\mu_2, \text{ and since } \mu_2 \geq 0 \Rightarrow s \leq 3.$$

 From $h_2 = x_1 = 0$ and $h_1 = x_2 - 4 = 0$ follows

$$x_2 = 4 \Rightarrow \mathbf{x}^* = [0, 4]^T.$$

- h_1 and h_3 active: $\mu_{1,3} \geq 0, \mu_2 = 0$

From (2a) follows

$$-1+\mu_1-\mu_3=0 \Rightarrow \mu_1=\mu_3+1,$$

$$1-s+2\mu_1=0 \Rightarrow s=1+2\mu_1=3+2\mu_3, \text{ and since } \mu_3 \geq 0 \Rightarrow s \geq 3.$$

 From $h_3 = x_2 = 0$ and $h_1 = 2x_1 - 4 = 0$ follows

$$x_1 = 2 \Rightarrow \mathbf{x}^* = [2, 0]^T.$$

Optimal solution

$$\mathbf{x}^* = \begin{cases} [0,4]^T, & \text{if } 0 \leq s < 3 \\ [x_1, 4 - 2x_1]^T, x_1 \in [0,2], & \text{if } s = 3 \\ [2,0]^T, & \text{if } 3 < s \leq 4 \end{cases}. \quad (3)$$

* c) Unconstrained problem with interior penalty function (6 P)

$$\min_{\mathbf{x} \in \mathbb{R}^2} P(\mathbf{x}, \tau) = f(\mathbf{x}) + \tau \sum_{i=1}^q \psi(h_i(\mathbf{x})) = \frac{1}{2}x_1 - x_2 - \tau \ln(x_1) - \tau \ln(2 - x_2) \quad (4)$$

Optimal solution in dependence of τ
Minimizing (4)

$$\nabla_{\mathbf{x}} P(\mathbf{x}, \tau) = \begin{bmatrix} \frac{1}{2} - \frac{\tau}{x_1} \\ -1 + \frac{\tau}{2-x_2} \end{bmatrix} = 0$$

results in

$$x_1^*(\tau) = 2\tau, \quad x_2^*(\tau) = 2 - \tau.$$

For $\tau \rightarrow 0$ the solution $\mathbf{x}^*$ approaches the optimal solution of the original problem.

d) Gradient method $\mathbf{x}^1 = \mathbf{x}^0 + \alpha^0 \mathbf{s}^0$ with search direction (6 P)

$$\mathbf{s}^0 = -\nabla_{\mathbf{x}} P(\mathbf{x}^0, \tau) = - \begin{bmatrix} \frac{1}{2} - \frac{\tau}{x_1} \\ -1 + \frac{\tau}{2-x_2} \end{bmatrix} = \begin{bmatrix} -\frac{1}{4} \\ \frac{1}{2} \end{bmatrix}$$

results in

$$\begin{aligned} x_1^1 &= 1 - \frac{1}{4}\alpha^0 \geq 0 \quad \Rightarrow \quad \alpha^0 \leq 4 \\ x_2^1 &= \frac{3}{2} + \frac{1}{2}\alpha^0 \leq 2 \quad \Rightarrow \quad \alpha^0 \leq 1 \end{aligned}$$

The maximum step size is $\alpha^0 = 1$, since for higher values the second constraint would be violated.
The next step of the gradient method results in

$$\mathbf{x}^1 = \begin{bmatrix} \frac{3}{4}, 2 \end{bmatrix}^T.$$

Task 2 (24 Points)

* a) Constraints and admissible set in Figure 1.

(3 P)

$$\nabla f(\mathbf{x}) = \begin{bmatrix} 4(x_1 - 5) \\ 2(x_2 - 2.5) \end{bmatrix}, \quad \nabla^2 f(\mathbf{x}) = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix} \geq 0 \quad (5)$$

The Problem is convex, since the cost function is convex and the admissible set is convex.

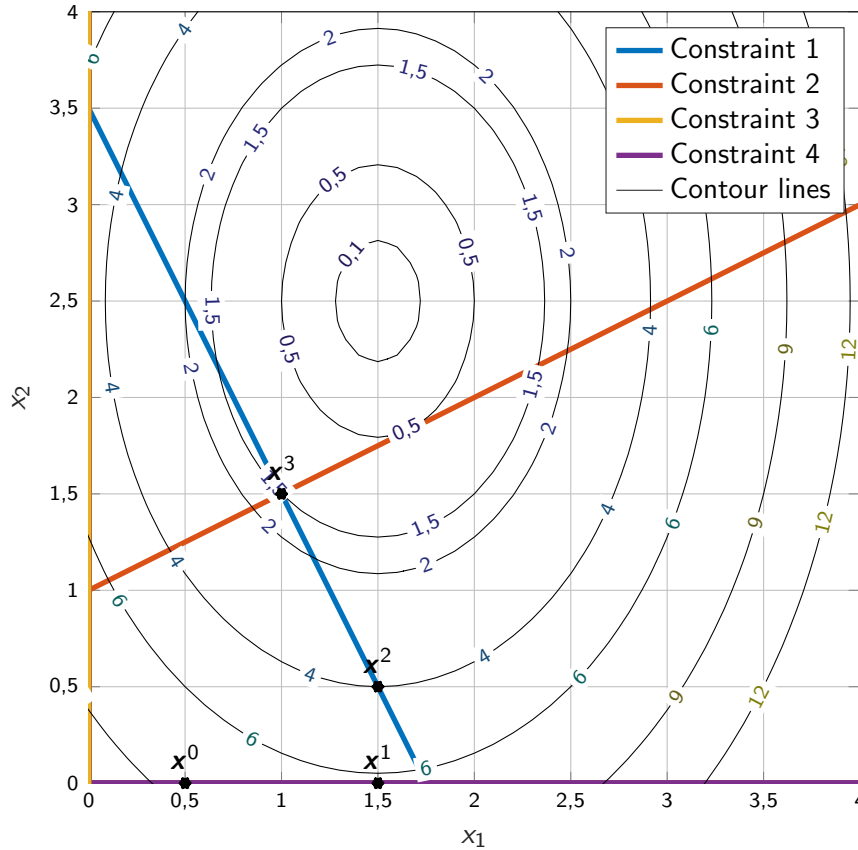


Figure 1: Constraints and admissible set for task 2a and iteration points from task 2c.

* b) Lagrangian function

(4 P)

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\mu}) = 2(x_1 - 1.5)^2 + (x_2 - 2.5)^2 + \boldsymbol{\mu}^T(\mathbf{C}\mathbf{x} - \mathbf{d}) \quad (6)$$

$$\text{with } \mathbf{C} = \begin{bmatrix} 2 & 1 \\ -0.5 & 1 \\ -1 & 0 \\ 0 & -1 \end{bmatrix} \text{ and } \mathbf{d} = [3.5, 1, 0, 0]^T.$$

Optimality conditions:

$$\begin{bmatrix} 4(x_1^* - 1.5) \\ 2(x_2^* - 2.5) \end{bmatrix} + \begin{bmatrix} 2 & -0.5 & -1 & 0 \\ 1 & 1 & 0 & -1 \end{bmatrix} \boldsymbol{\mu}^* = \begin{bmatrix} 4(x_1^* - 1.5) + 2\mu_1^* - \frac{1}{2}\mu_2^* - \mu_3^* \\ 2(x_2^* - 2.5) + \mu_1^* + \mu_2^* - \mu_4^* \end{bmatrix} = \mathbf{0} \quad (7a)$$

$$\mathbf{C}\mathbf{x}^* - \mathbf{d} \leq \mathbf{0} \quad (7b)$$

$$\boldsymbol{\mu}^* \geq \mathbf{0} \quad (7c)$$

$$\boldsymbol{\mu}^{*\top}(\mathbf{C}\mathbf{x}^* - \mathbf{d}) = 0. \quad (7d)$$

c) Solution with active set method

(10 P)

- $\mathbf{x}^0 = [0.5, 0]^T \Rightarrow \mathcal{A} = \{4\}$.
- $\mathbf{x}^1 = [1.5, 0]^T$ (graphical solution). Computation of the multipliers from (7a) results in $\mu_4 = -5$, thus remove 4 $\Rightarrow \mathcal{A} = \{\}$.
- $\mathbf{x}^2 = [1.5, 0.5]^T$ (graphical solution). Constraint 1 gets active $\Rightarrow \mathcal{A} = \{1\}$.
- $\mathbf{x}^3 = [1.0, 1.5]^T$ (graphical solution). Constraint 2 gets active $\Rightarrow \mathcal{A} = \{1, 2\}$. Computation of the multipliers from (7a)

$$\begin{bmatrix} -2 + 2\mu_1^* - \frac{1}{2}\mu_2^* \\ -2 + \mu_1^* + \mu_2^* \end{bmatrix} = \mathbf{0} \quad (8)$$

results in $\mu_1 = 2 - \mu_2$ and $\mu_2 = \frac{4}{5} \geq 0$, thus $\mu_1 = \frac{6}{5} \geq 0$ and $\mathbf{x}^3 = \mathbf{x}^*$.

d) The Lagrangian multipliers at the optimal point represent a measure of the sensitivity of the cost function with regard to a relaxation of the corresponding constraint. That is, the larger the respective multiplier, the more the costs can be reduced by relaxing the corresponding constraint. Here, relaxing h_1 would lead to a smaller optimal solution than relaxing h_2 , since $\mu_1 > \mu_2$. (2 P)

e) Graphical solution: $\mathbf{x}^ = [0.5, 2.5]^T$. The problem is still convex, since the cost function as well as the admissible set are convex. The active set method is no longer suitable, since the quadratic constraint would no longer stay active along a search direction $\mathbf{s}^k$. (5 P)

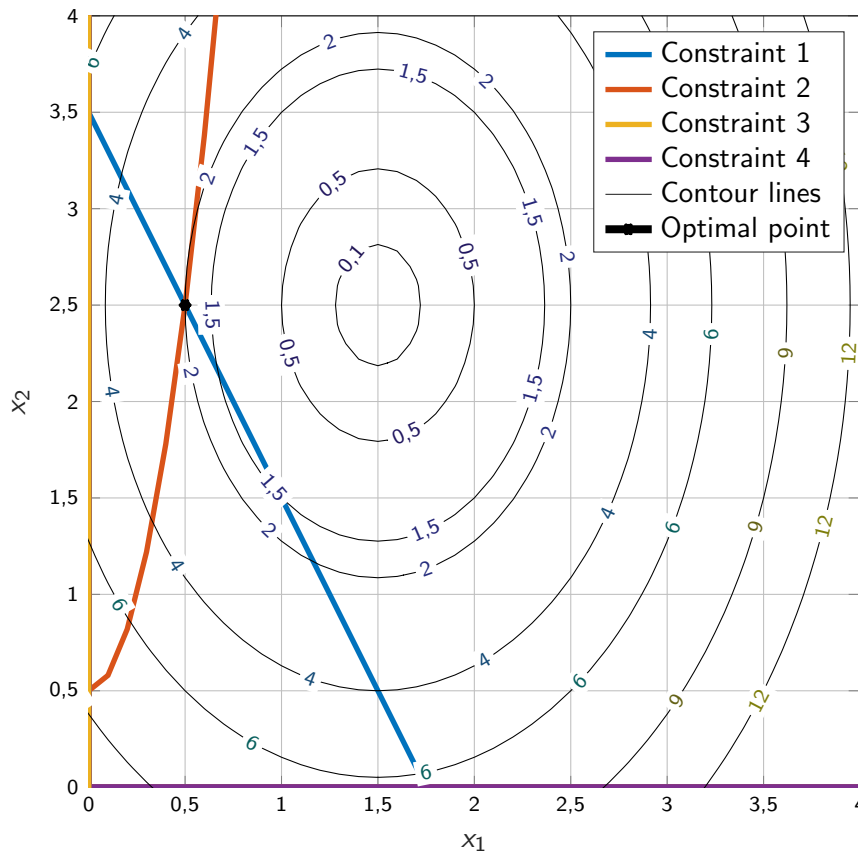


Figure 2: Admissible set with new constraint and optimal point from task 2e.

Task 3 (20 Points)

* a) Hamiltonian function

(4 P)

$$H(x, u, \lambda) = l(x, u) + \lambda f(x, u) = ax + \frac{1}{2}u^2 + \lambda u \quad (9)$$

Optimality conditions

$$\dot{x} = \frac{\partial H}{\partial \lambda} = u, \quad x(0) = 1, \quad x(t_f) = 0 \quad (10a)$$

$$\dot{\lambda} = -\frac{\partial H}{\partial x} = -a \quad (10b)$$

$$0 = \frac{\partial H}{\partial u} = u + \lambda \quad (10c)$$

$$H(x(t_f), u(t_f), \lambda(t_f), t_f) = -\frac{\partial V}{\partial t}(x(t_f)) = 0. \quad (10d)$$

b) Computation of the optimal solution

(10 P)

- From (10b) follows

$$\lambda(t) = -at + c_1. \quad (11)$$

Thus for (10c) and (10a) follows

$$u(t) = -\lambda(t) = at - c_1 \quad (12)$$

$$\dot{x}(t) = u(t) = at - c_1 \Rightarrow x(t) = \frac{a}{2}t^2 - c_1 t + c_2. \quad (13)$$

- Initial condition $x(0) = 1$ leads to $c_2 = 1$ and

$$x(t) = \frac{a}{2}t^2 - c_1 t + 1. \quad (14)$$

- The end time t_f and constant c_1 are calculated from (10d)

$$H(x(t_f), u(t_f), \lambda(t_f), t_f) = ax(t_f) + \frac{1}{2}u(t_f)^2 + \lambda(t_f)u(t_f), \quad (15)$$

with terminal condition $x(t_f) = 0$ and $\lambda(t) = -u(t)$ from (10c) follows

$$H(x(t_f), u(t_f), \lambda(t_f), t_f) = \frac{1}{2}u(t_f)^2 - u(t_f)^2 = -\frac{1}{2}u(t_f)^2 \stackrel{!}{=} 0, \quad (16)$$

thus $u(t_f) \stackrel{!}{=} 0$ and

$$u(t_f) = at_f - c_1 \stackrel{!}{=} 0 \Rightarrow t_f = \frac{c_1}{a}. \quad (17)$$

With terminal condition $x(t_f) = 0$ follows

$$\begin{aligned} x(t_f) &= \frac{a}{2}t_f^2 - c_1 t_f + 1 = \frac{a}{2}\left(\frac{c_1}{a}\right)^2 - c_1 \frac{c_1}{a} + 1 = -\frac{c_1^2}{a} + 1 \stackrel{!}{=} 0 \\ \Rightarrow c_1 &= \pm\sqrt{2a}, t_f = \pm\frac{1}{a}\sqrt{2a} \end{aligned} \quad (18)$$

Since $a > 0$ and the end time $t_f \geq 0$, it follows $t_f = \sqrt{\frac{2}{a}}$ and $c_1 = \sqrt{2a}$.

$$u^*(t) = at - \sqrt{2a} = -\lambda^*(t) \quad (19a)$$

$$x^*(t) = \frac{a}{2}t^2 - \sqrt{2a}t + 1 \quad (19b)$$

$$t_f^* = \sqrt{\frac{2}{a}}. \quad (19c)$$

* c) Effect of a on solution and end time: larger a , leads to faster decay of u^* , x^* to zero and the optimal final end gets shorter. (2 P)

* d) Consideration of input constraints (Pontryagin's max. principle) (4 P)

$$u^* = \arg \min_{u \in [-1,1]} H = \arg \min_{u \in [-1,1]} \frac{1}{2}u^2 + \lambda u \quad (20)$$

Calculate minimum

$$\frac{\partial H}{\partial u} = u + \lambda \stackrel{!}{=} 0 \quad \Rightarrow \quad u^0 = -\lambda \quad (21)$$

$$\Rightarrow u^* = \begin{cases} -1, & \text{if } u^0 < -1 \\ u^0, & \text{if } u^0 \in [-1, 1] \\ 1, & \text{if } u^0 > 1 \end{cases} \quad (22)$$

Task 4 (20 Points)

- * a) • Optimal solution $x^* = \frac{1}{2}$ from the equality constraint (6 P)

$$2x - 1 = 0 \Rightarrow x = \frac{1}{2}. \quad (23)$$

The optimal λ^* follows from the Lagrangian

$$L(x, \lambda) = (x - 2)^2 + \lambda(2x - 1) \quad (24)$$

and the first optimality condition

$$\nabla_x L(x, \lambda) = 2(x - 2) + 2\lambda = 0 \Rightarrow \lambda^* = \frac{3}{2} \quad (25)$$

- Augmented Lagrangian function

$$L_c(x, \lambda) = (x - 2)^2 + \lambda(2x - 1) + \frac{c}{2}(2x - 1)^2 \quad (26)$$

- Calculate x^k

$$\nabla_x L_c(x, \lambda) = 2(x - 2) + 2\lambda + 2c(2x - 1) = (2 + 4c)x - 4 + 2\lambda - 2c = 0 \quad (27)$$

$$\Rightarrow x^k = \frac{c^k + 2 - \lambda^k}{1 + 2c^k} \quad (28)$$

- * b) Convergence of λ^{k+1} (4 P)

$$\lambda^{k+1} = \lambda^k + c^k(2x^k - 1) = \lambda^k + c^k\left(2\frac{c^k + 2 - \lambda^k}{1 + 2c^k} - 1\right) = \lambda^k + c^k\frac{3 - 2\lambda^k}{1 + 2c^k} \quad (29)$$

$$\lambda^{k+1} - \lambda^* = \lambda^k - \lambda^* + c^k\frac{3 - 2\lambda^k}{1 + 2c^k} = \frac{\lambda^k + 2c^k\lambda^k - \lambda^* - 2c^k\lambda^* + 3c^k - 2\lambda^k c^k}{1 + 2c^k} \quad (30a)$$

$$= \frac{\lambda^k - \lambda^*}{1 + 2c^k} \Rightarrow C = \frac{1}{1 + 2c^k}. \quad (30b)$$

The augmented Lagrangian method converges for $C < 1$, thus $1 + 2c^k > 1$ leads to $c^k > 0$.

Alternative solution with $x^k = \frac{c^k - \lambda^k + 2}{2 + 2c^k}$ and $\lambda^* = 1$:

$$\lambda^{k+1} = \lambda^k + c^k\left(2\frac{c^k - \lambda^k + 2}{2 + 2c^k} - 1\right) = \lambda^k + c^k\frac{1 - \lambda^k}{1 + c^k} \quad (31)$$

$$\lambda^{k+1} - \lambda^* = \lambda^k - \lambda^* + c^k\frac{1 - \lambda^k}{1 + c^k} = \frac{\lambda^k + c^k\lambda^k - \lambda^* - c^k\lambda^* + c^k - \lambda^k c^k}{1 + c^k} \quad (32a)$$

$$= \frac{\lambda^k - \lambda^*}{1 + c^k} \Rightarrow C = \frac{1}{1 + c^k}. \quad (32b)$$

The augmented Lagrangian method converges for $C < 1$, thus $1 + c^k > 1$ leads to $c^k > 0$.

* c) CLF inequality

(6 P)

$$\dot{V}(x) + l(x, -kx) = 2px\dot{x} + qx^2 + r(-kx)^2 \quad (33a)$$

$$= 2px(ax - bk) + qx^2 + r\left(-\frac{bp}{r}x\right)^2 \quad (33b)$$

$$= 2pax^2 - 2pb\frac{pb}{r}x^2 + qx^2 + \frac{(bp)^2}{r}x^2 \quad (33c)$$

$$= 2pax^2 - \frac{b^2p^2}{r}x^2 + qx^2 \quad (33d)$$

$$= \left(2pa - \frac{b^2p^2}{r} + q\right)x^2 = 0 \quad \forall x \in \mathbb{R} \quad (\text{alg. Riccati}) \quad (33e)$$

and $S_\beta = \mathbb{R}$.

* d) Boundary conditions for the state

(4 P)

$$x_i(t_0) = x_{0,i}, \quad i \in \mathcal{I}_0 \quad (34a)$$

$$x_j(t_f) = x_{0,j}, \quad j \in \mathcal{I}_f \quad (34b)$$

Boundary conditions for the adjoint state

$$\lambda_i(t_0) = 0, \quad i \notin \mathcal{I}_0 \quad (35a)$$

$$\lambda_j(t_f) = \frac{\partial V}{\partial x_j}(\mathbf{x}(t_f), t_f), \quad j \notin \mathcal{I}_f \quad (35b)$$